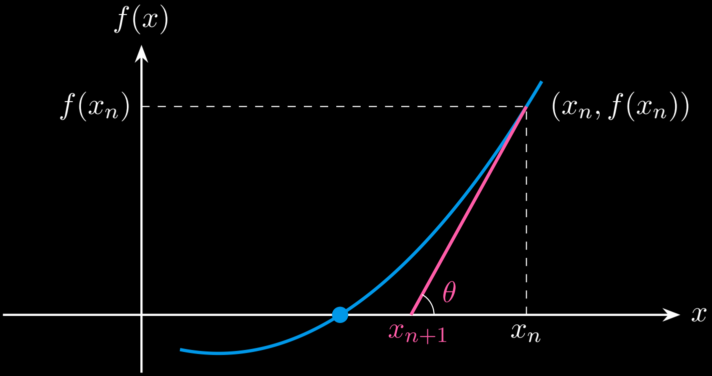
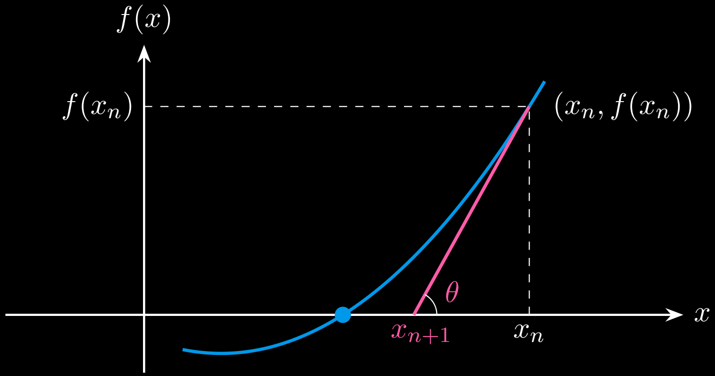
Intuitive Understanding of Newton's Method

$$\text{slope} = \frac{\text{change in } y}{\text{change in } x} = \frac{0 - f(x_n)}{x_{n+1} - x_n} = f'(x_n)$$



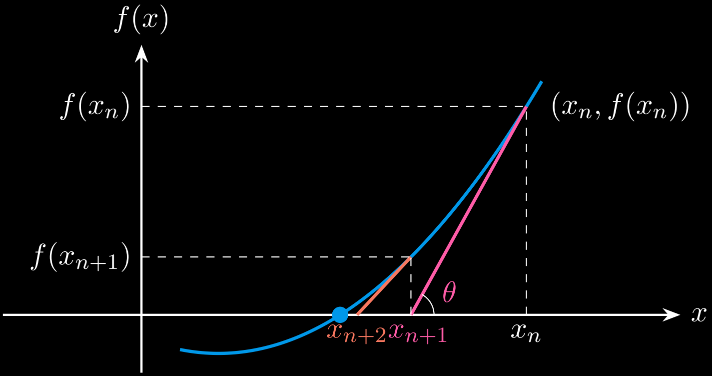
Intuitive Understanding of Newton's Method

$$x_{n+1} - x_n = -\frac{f(x_n)}{f'(x_n)} \quad \Rightarrow \quad x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$



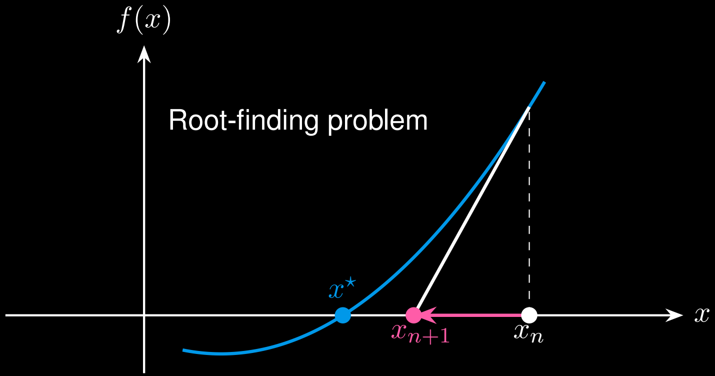
Intuitive Understanding of Newton's Method

$$x_{n+2} = x_{n+1} - \frac{f(x_{n+1})}{f'(x_{n+1})}$$



Intuitive Understanding of Newton's Method

$$x := x - \frac{f(x)}{f'(x)}$$ (update x until convergence)



Intuitive Understanding of Newton's Method

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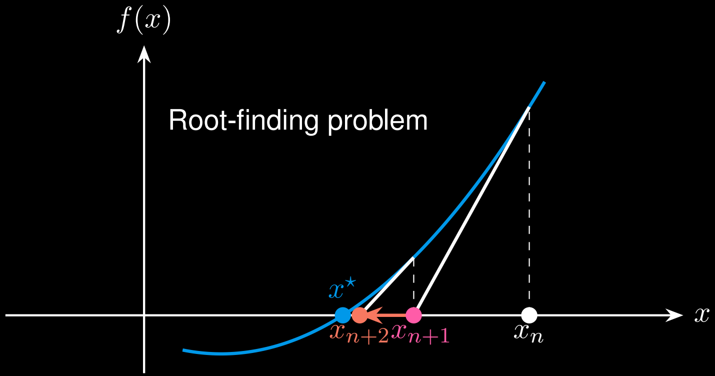
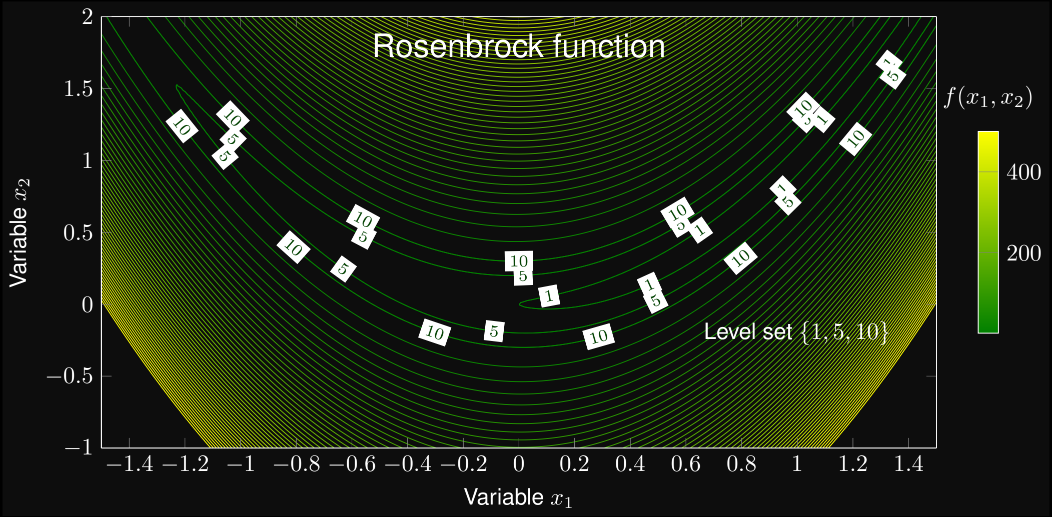


Unconstrained Nonlinear Optimization

Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$



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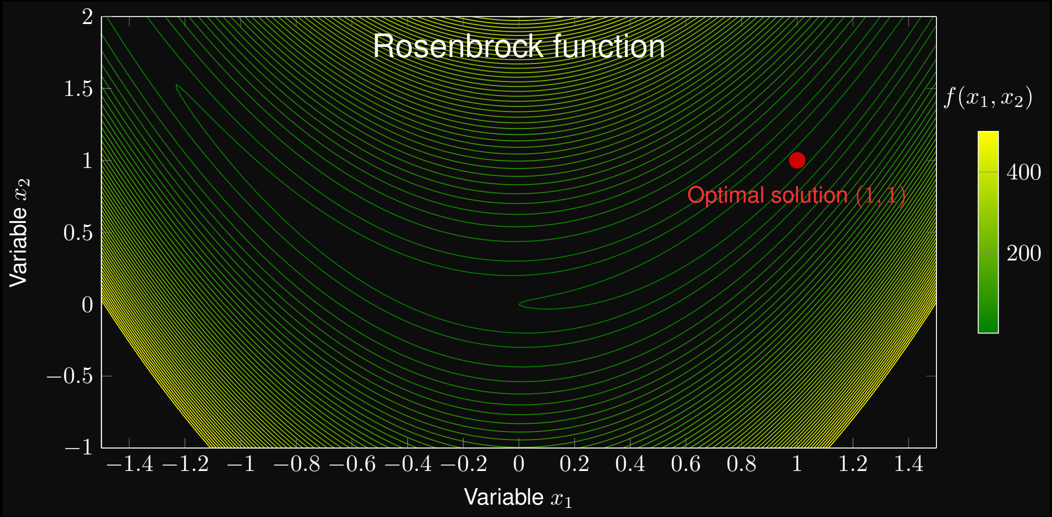

First- and Second-Order Derivatives

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$

Unconstrained Nonlinear Optimization

The optimization can be converted into a root finding problem:

$$\min_{x_1, x_2} \quad f(x_1, x_2) \qquad \Leftrightarrow \qquad \text{Find} \quad \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix} = \mathbf{0} \quad (\text{gradient} = \mathbf{0})$$

Newton's method for root finding:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} := \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \underbrace{\begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} \end{bmatrix}^{-1}}_{\text{inverse of Hessian matrix}} \underbrace{\begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}}_{\text{gradient}}$$

Unconstrained Nonlinear Optimization

$$\min_{x_1, x_2} \quad f(x_1, x_2) = (1 - x_1)^2 + 100(x_2 - x_1^2)^2$$